FIFA Data Hub

Legend

Input & output

Module

Script/Notebook

main_function
Main function to compute events

routine
subroutine 1
subroutine 2
Routine with subroutines

Flow of information

Raw tracking data
Raw positions
Metadata

Module tracking_routines
- *get_halves_tracab*
- *parse_half_tracab*
- *get_active_goal*
- *prepare_tracking_data*
  - *trim_tracking_data*
  - *filter_position (optional)*
  - *compute_velocities*

Tracking dataframe
Player (x, y, vx, vy)
Ball (x, y, z, vx, vy, in-play/dead)
Pitch dimensions

Game ID
User-defined parameters

Module set_inputs
- *parse_lineups*
- *get_params*
- *load_game_params*

Game metadata (IFES)
player IDs to teamID & jersey number
Goalkeeper IDs

Parameters
See ReadMe file

Module game_time

Module utils

Module football_logic

possession_model
- Extract possession/pass/contest information using player proximity to ball
- Extract dead ball intervals & potential set pieces based on spatial player configurations

assign_possession_contest
- *ball_direction*

set_saves_shots
- *count_player_in_penalty_box*
- *pitch_zone*
- *in_penalty_box*
- *ball_direction*
- *match_player_interval*
- *shot_on_target*

extract_dead_ball_segments
- *extract_penalty_trigger*
  penalty_shooter
  between_goal_pkspot
  in_own_goal_line
  penalty_trigger
- *extract_kickoff_trigger*
  kickoff_average_y_10yards
- *extract_corner_trigger*
  corner_trigger
- *extract_throwin_trigger*
  throwin_trigger
- *extract_goalkick_trigger*
  goalkick_trigger
  check_ball_in_goal_2d
- *get_index_from_trigger*
- *isin_complete_trigger*
- *isin_incomplete_trigger*
- *check_cornerkick*
- *check_throwin*
- *check_goalkick*

set_gains_losses
- *in_pitch_player*
- *match_player_interval*

event_detection
- Detect possession gains, passes and continued possessions
- Detect set pieces
- Detect saves & shots
- Detect score
- Separate ball control & events
- Apply temporal logic to events (optional)
- Detect duels
- Detect team possession
- Add possession logic to events (optional)
- Add contextual attributes to events (optional)
- Detect pressure (optional)
- Detect defensive modes (optional)
- Apply pitch control (optional)

set_score
- *ball_direction*
- *pitch_zone*
- *shot_on_target*
- *in_penalty_box*

set_event_ballcontrol

set_event_logic

set_duels

set_team_possession

set_possession_logic
- *count_player_in_penalty_box*
- *ball_direction*
- *pitch_zone*
- *in_cross_box*

set_pressure
- *in_penalty_box*
- *create_team_pressure*

set_dead_ball_events
- *check_ball_in_goal_2d*
- *check_ball_in_goal_z*
- *new_sp_event*
- *get_displacement_player_frames*
- *goalkick_trigger*
- *throwin_trigger*
- *corner trigger*

postprocess
- *in_hull*
- *compute_angle*

set_defensive_modes
- *raw_defensive_mode*
  is_high_press
  is_gk_press
  is_mid_block
  is_low_block
  is_deep_low_block
  is_advanced_low_block
- *collapse_short_intervals*

Utility Functions
- *names*
- *jersey_team*
- *rolles*
- *events_to_json*
- *unroll_tuples*

Module pitch_control

apply_pitch_control
- ***Module** pitch_control*
- *in_cone_vision*

events

Notebook
local_autoeventer.ipynb

Script gen_video.py

Module graphics

WholePitchVideo
- _draw_football_field
- _animate
- get_anim

tracking data video with events overlaid